



```
model = yolov8s-face-lindevs.pt
format = ncnn
# or
yolo export model = yolov8s-face-
lindevs.pt format = ncnn imgsz = 320
# or 640
```

The resulting NCNN model is then used directly for real-time face detection. The main tracking program on the laptop is launched using the caffeinate utility to prevent the system from entering sleep mode: *caffeinate -i python your_python_code.py*

Inside the script, the external USB camera is opened automatically, and NCNN inference runs continuously. The centroid of each detected face is calculated, and short packets containing the pan angle, tilt angle, and face count are transmitted to the ESP32 via ESPConnect. At the same time, the VL53L0X distance reading is received from the ESP32 and overlaid on the live video feed.

On the ESP32, the firmware first initialises the shared I²C bus, PCA9685 PWM driver, VL53L0X distance sensor, and OLED display. It then enters a continuous loop that listens for command packets from the laptop, updates the corresponding servo pulse widths through the PCA9685, reads the current distance from the VL53L0X, and refreshes the OLED with the latest face count, distance, and servo angles. Since the ESP32 executes commands more slowly than a direct Raspberry Pi-Hailo configuration, the raster-scan pattern described earlier is used to systematically cover the field of view. This approach provides more reliable target acquisition than a zig-zag scan.

Construction and testing

The system is assembled on a perforated prototype board, with the ESP32, PCA9685 PWM driver, VL53L0X ToF distance sensor, and 0.96-inch (2.43cm) OLED display interconnected through the I²C bus. Two servo motors form the pan-tilt

mechanism, with the USB camera mounted on the upper platform. A regulated 5V supply powers the control circuitry and servos, while the laptop runs the YOLOE/YOLOv8-face detection software and communicates with the ESP32 through ESPConnect. The real-time face tracking at a frame rate exceeding 50 FPS is shown in Fig. 5.

For testing, power is first applied to the system, and the OLED splash screen and VL53L0X distance readings are verified. The Python tracking program is then launched on the laptop using the caffeinate command, and the USB camera is selected as the active video source. A face or prompted object is placed within the camera's field of view. Detected faces should appear promptly with bounding boxes, while the video overlay displays the face count, distance, and frame rate.

The pan-tilt mechanism performs a raster scan until faces are detected, after which it tracks the group centroid and re-centres when the target position changes. The OLED simultaneously displays the detected face count, distance, and servo angles.

The system was tested with a group scene and successfully detected 24 faces, as shown in Fig. 5. The measured distance was approximately 157mm, while the laptop achieved about 59 FPS under the test conditions. The subject was then moved across the field of view to verify smooth servo tracking. Any minor oscillation can be reduced by lowering the proportional gain or introducing a small dead-band in the ESP32 firmware.

These tests confirm the operation of the face-detection, distance-sensing, pan-tilt tracking, and display functions of the laptop-ESP32 architecture.

Applications and future scope

The completed system has several practical applications. It can serve as an intelligent surveillance platform

for tracking people or objects and detecting events such as restricted-area entry or specific objects being carried. In meeting rooms, it can provide automatic camera framing by continuously tracking the group centroid. For elderly or patient care, it can monitor movement and, when combined with fall detection, provide an automated safety system.

YOLOE's prompt-based recognition also enables low-cost human-robot interaction by identifying gestures, tools, or other objects and triggering actions through the ESP32. In retail, it can track customers, count visitors, and monitor product interactions. It can also track personal or restricted objects and generate alerts when they are moved. As an educational platform, the system demonstrates real-time AI inference, embedded control, and communication in an easily expandable system. It can further be adapted for crowd monitoring and vision-based smart-home automation.

With additional servos and actuators, the platform can evolve into advanced robotic systems, such as a multi-axis pick-and-place arm or an intelligent object-retrieval system. Gesture recognition could also enable touchless robotic control for assistive applications.

Overall, the system demonstrates how prompt-driven YOLOE vision can reduce the need for conventional model retraining, while the laptop-ESP32 architecture provides responsive tracking and hardware control. The architecture can also be extended to other AI-driven automation and assistive applications. **EFY**

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